Predicting School Choice in Gujarat: A Survey-Weighted Logistic Regression in R

What this project does

This project builds a statistical model that estimates the probability of a child enrolling in a private school versus a public school, based on their household's characteristics. The goal is to profile which types of students end up in which school system in Gujarat, India. This statistical model is a part of the paper titled - "Transforming School Education in Gujarat: A Governance Perspective", authored by Mayur Matang under the supervision of Prof. Hans Boehmer and Prof. Sarah Dickerson at Duke University, USA.

Data and Method

The analysis uses microdata from the National Sample Survey Organization's Education Consumption Survey, 2018 – one of India's largest household surveys – covering ~4,000 children in Gujarat. Because the outcome is a yes/no question (private school or not), I used a logistic regression model, which predicts a probability rather than a number. The model respects the survey's complex design: it applies the survey's sampling weights (via R's survey package), so results represent Gujarat's actual population rather than just the sampled households. The model fits the data well – McFadden's pseudo-R^2 is 0.71 (scale of 0 to 1, higher is better).

Key findings

- **Age**: Younger students are more likely to enroll in private schools than older ones – the odds fall steadily as age rises, possibly because primary education costs less than secondary education.
- **Social group**: Compared to the privileged group, the odds of private school enrollment drop by 91% for Tribes, 77% for Scheduled Castes, and 68% for Other Backward Classes.
- **Religion**: Religious minorities are 65% less likely to enroll in private schools. Together with the social group results, this points to inequities the ministry must address.
- **Distance to school**: The likelihood of private school enrollment rises with distance – a sign that private schools have low penetration in Gujarat.
- **Location**: Rural students are 75% less likely to enroll in private schools than urban students, possibly because public schools crowd out private ones in rural areas, or because higher rural poverty makes running a private school unviable.
- **Private coaching**: Students taking private coaching are 53% less likely to be in private schools, supporting the perception that public school students supplement their learning with coaching, likely a cheaper option than private school fees.
- **Not significant**: Household size doesn't predict school type. More surprisingly, neither does gender. It is generally believed that boys receive preferential treatment within

households and are admitted to private schools while girls are not, but the data show no such gap once other factors are accounted for.

Tools

Built entirely in R using packages: tidyverse (data wrangling), survey (weighted estimation), haven and labelled (importing labelled survey microdata), margins and jtools (model interpretation and output), gt/gtsummary (publication-quality tables), here (reproducible file paths).

Logit Model: Odds Ratio

Characteristic	OR[1]	SE[1]	95% CI[1]	p-value
Age_group				
3-5 yrs	—	—	—	
6-10 yrs	0.94	0.419	0.41, 2.14	0.9
11-15 yrs	0.65	0.428	0.28, 1.52	0.3
16-20 yrs	0.30	0.463	0.12, 0.74	0.009
above 20 yrs	0.08	0.594	0.02, 0.24	<0.001
Gender				
Male	—	—	—	
Female	1.15	0.156	0.84, 1.56	0.4
Social_group				
Others	—	—	—	
Scheduled Tribes	0.09	0.482	0.04, 0.24	<0.001
Scheduled Castes	0.23	0.491	0.09, 0.60	0.003
Other Backward Classes	0.32	0.233	0.20, 0.51	<0.001
Religion_new				
Majority	—	—	—	
Minority	0.35	0.431	0.15, 0.83	0.016
Household_size	1.02	0.051	0.92, 1.13	0.7
Residance_insttn_distance				
<1 km	—	—	—	
1 km d < 2 kms	3.74	0.344	1.90, 7.35	<0.001
2 km d < 3 kms	4.52	0.379	2.14, 9.52	<0.001
3 kms d < 5 kms	7.66	0.390	3.56, 16.5	<0.001
>= 5 kms	9.53	0.399	4.35, 20.9	<0.001
Sector				
urban	—	—	—	
rural	0.25	0.297	0.14, 0.45	<0.001
Taking_pvt_coaching				
No	—	—	—	
Yes	0.47	0.305	0.26, 0.85	0.012

[1] OR = Odds Ratio, SE = Standard Error, CI = Confidence Interval

Source: NSSO Education Consumption 2018.

NSSO data is filtered for school students in Gujarat state.